Unit 1: Worked solutions

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

E01

$G_1 = 2/0.1 = 20\,\mu S$ and $G_2 = 6/0.2 = 30\,\mu S$. The different ratios exclude a fixed-conductance, origin-passing linear model over these points; they do not uniquely identify a conduction mechanism. Record bias and polarity, pulse duration and timing, temperature, programmed state, compliance and instrument conditions.

E02

SET increases conductance; RESET decreases it. Forming is a device-dependent initialization, and read measures the state while requiring a disturbance check. Under the stated bipolar convention RESET normally has negative polarity. Different stacks need not share this sign. Current compliance limits excessive conduction and heating, affects the final state, and does not guarantee immunity from failure.

E03

No: hysteresis does not uniquely identify a mechanism. ECM commonly involves mobile metal cations and redox; VCM involves oxygen ions or vacancies and local valence changes. Electrode substitution, isotope tracing, local chemistry or in-situ microscopy can provide evidence. Control field, electrodes and environment, and account for limited spatial resolution, sample preparation and probe-induced changes; correlation alone need not establish causation.

E04

Column 1 branches carry $2(0.2) = 0.4$ and $1(0.1) = 0.1\,\mu A$, totaling $0.5\,\mu A$. Column 2 branches carry $1(0.2) = 0.2$ and $3(0.1) = 0.3\,\mu A$, also totaling $0.5\,\mu A$. At the stated actual second-row crosspoint voltage, the totals become $0.4 + 0.08 = 0.48\,\mu A$ and $0.2 + 0.24 = 0.44\,\mu A$. Linear branch equations remain valid using actual local voltages, but the input terminal voltage 0.1 V cannot be substituted at every second-row crosspoint. Generally wire resistance makes local voltages unequal, requiring a circuit solution.

E05

The defined margin is $5 - 3 = 2\,\mu A$. With the stated strict decision rule, every $3 \le T < 5\,\mu A$ separates the intervals. Other equality conventions change endpoint inclusion. Overlap precludes a guarantee of zero error from the ranges alone; error rate needs state distributions, prior probabilities, threshold and measurement noise.

Unit 1: Worked solutions (continued)

E06

Report programming amplitude and duration, current compliance, read voltage and timing, temperature, failure criterion, sample count and structure. Cycle-to-cycle variability describes repeated operations on one device; device-to-device variability compares devices. A single accelerated-test point does not observe ten years directly. Extrapolation needs a validated failure model, fitted parameters and uncertainty, and must consider possible changes of failure mechanism.

E07

For constant P, $T(t) = 300 + P(10^5)(1 - e^{-t/(10\,\mathrm{ns})})$. The separate 4 mW, 30 ns trial gives $300 + 400(1 - e^{-3}) = 680.1$ K; the separate 10 mW, 10 ns trial gives $300 + 1000(1 - e^{-1}) = 932.1$ K. After the second pulse, set $P = 0$: $T(t) = 300 + 632.1e^{-t/(10\,\mathrm{ns})}$. At 600 K, $t = 10\ln(632.1/300) = 7.45$ ns. The first peak is above the illustrative 600 K threshold and below 900 K; the second exceeds 900 K. Neither crossing suffices to establish a phase: nucleation, growth, spatial temperature, cooling rate, prior phase and subsequent retention must be assessed.

E08

The conductance sums for inputs 00, 01, 10, 11 are 2, 11, 11, 20 µS, giving currents 0.2, 1.1, 1.1, 2.0 µA. Applying the strict $I > 0.6\,\mu$A comparator gives 0, 1, 1, 1: OR. Initialize an independent output cell to HRS; a digital controller applies a SET pulse if and only if the comparator outputs 1, then verifies the result. The comparator, controller and programming path are required; the summed current alone does not rewrite the output. The calculation assumes ideal non-disturbing reads and no variations.

E09

Immediately after the first pulse $g(0^+) = 2\,\mu$S. Exponential decay gives $g(10^-) = 2e^{-1} = 0.7358\,\mu$S; adding the second spike gives $g(10^+) = 2.7358\,\mu$S. With no further pulse, $g(20) = 2.7358e^{-1} = 1.0064\,\mu$S. The stated 0.1 V read produces $I = 0.1 \times 2.7358 = 0.27358\,\mu$A. This is a hypothetical linear, additive and exponential model, not a universal device law. Saturation, nonlinear relaxation, history and read disturbance require independent measurements. Millisecond-scale response does not establish long-term nonvolatile retention; that requires a separate zero-bias waiting test with stated temperature, read protocol and failure criterion.

E10

For $P(1) = 0.6$, $H_{min} = -\log_2(\max(0.6, 0.4)) = 0.737$ bit per raw bit. Independence gives $P(01) = 0.4(0.6) = 0.24$ and $P(10) = 0.6(0.4) = 0.24$, hence $P(\mathrm{accepted}) = 0.48$ per pair. Conditional on acceptance, each output is equally likely; average yield is $0.48/2 = 0.24$ output bits per raw bit. A marginal bias does not determine temporal dependence or entropy rate. Correlations and nonstationarity can also invalidate the debiasing argument, while finite test success does not prove unpredictability against an adversary.

Unit 1: Worked solutions (continued)

E11

Let conductances be in µS. By KCL, $v = (0.5G_P + G_Q)/(G_P + G_Q + 3)$ and $V_Q^{dev} = 1 - v$. For input 00: $v = 1.5/5 = 0.3000$ V and $V_Q^{dev} = 0.7000$ V, so Q SETs to 1. For 01: $v = 10.5/14 = 0.7500$ V and $V_Q^{dev} = 0.2500$ V, so Q remains 1. For 10: $v = 6/14 = 0.4286$ V and $V_Q^{dev} = 0.5714$ V, below the assumed SET threshold, so Q remains 0. For 11: $v = 15/23 = 0.6522$ V and $V_Q^{dev} = 0.3478$ V, so Q remains 1. The output is 1, 1, 0, 1: $\neg p \vee q$.

For the inputs 00, 01, 10 and 11, respectively, $V_P^{dev} = 0.5 - v$ is 0.2000, -0.2500, 0.0714 and -0.1522 V. Initially $|V_P^{dev}|$ is at most 0.25 V, within the assumed non-disturb window. After 00 switches to $G_Q = 10\,\mu$S, $v = 10.5/14 = 0.7500$ V and $V_Q^{dev} = 0.2500$ V; the assumed latch retains Q in LRS. Switching speed, actual transient voltages, stochastic thresholds, retention, pulse width and device-to-device distributions can invalidate the simplified circuit assumptions.

E12

Independence within each pair gives $P(Y_i = 1) = P(A_i = 1)P(B_i = 1) = 0.6(0.5) = 0.3$. For 100 independent identically distributed pairs, $S = \sum_i Y_i$ is binomial with $E[S] = 100(0.3) = 30$ and $\mathrm{sd}(S) = \sqrt{100(0.3)(0.7)} = \sqrt{21} \approx 4.58$. As a counterexample, let $A_i = B_i = X_i$ for a fair bit. Both marginals equal 0.5, yet $P(A_i \wedge B_i = 1) = 0.5$, not 0.25. If Y_i are correlated across time, the variance is $\sum_i \mathrm{Var}(Y_i) + 2\sum_{i<j} \mathrm{Cov}(Y_i, Y_j)$ and need not equal the binomial variance.

Lin Cing https://github.com/xinhi6716/resistive-memory-self-study